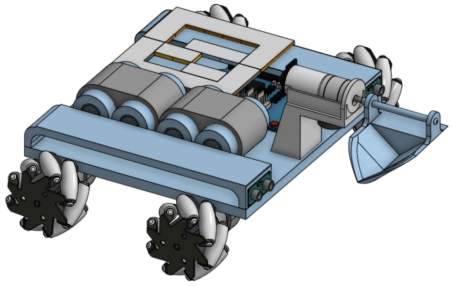
Constructing a Prototype

The first step in designing our prototype was coming up with different ideas on how to solve the problem of our mission. We first thought of using a scooper to do both mission actions: picking up and planting. What would happen is that the scooper would disturb the plantable substrate, orzo, and then the seed would be in the back of the scooper. So as the scooper did a 180-degree turn, the seed would fall out into a small hole the scooper carved out. Then, when the scooper went back 180 degrees, it would drop the orzo over the seed, successfully planting it. Then, for the collection of the nonplantable substrate, it would just pick up the rocks and hold them till the end. We also decided on using 4 mecanum wheels, two on each side of our robot, to allow our OTV to move from side to side without having to make turns to dodge obstacles. We also needed something to detect which plots were plantable and which were the nonplantable ones. Once milestone 1 was achieved, we had a decent idea about what we were doing and how we were going to do it.

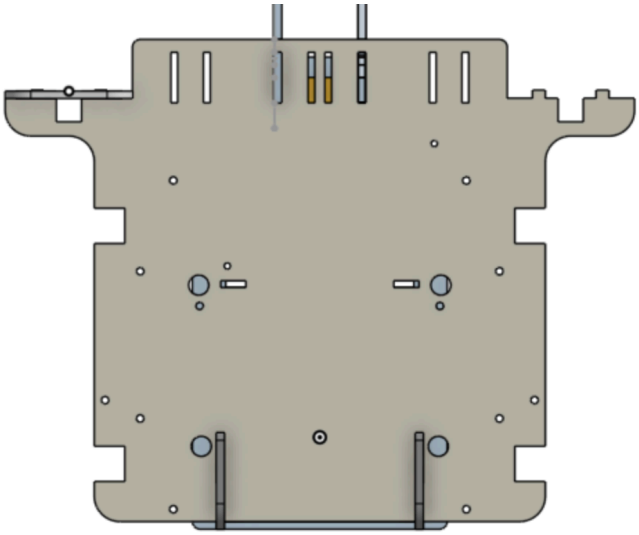


[1] Our Milestone 2 CAD

We had our first design, which was disorganized and did not have many electronic parts connected to it. It had everything that we envisioned, with a scooper, wheels, and a second layer where our Aruco marker would lie. We continued to add electronics to our design, and soon we started to have a concrete idea of how our robot was going to look and move. By reviewing UMD documents online, we found the intention for seed planting was to use an ESP32 wifi module and a camera connected with the Jetson's ML model to differentiate pebbles from orzo. Approximately halfway through the semester, our mission was changed, and we were unable to continue without the original idea of the scooper. The rocks were switched to be coarser and were unable to be picked up by the scooper. Alongside this, the plot sizes were changed to be 22% smaller than before, making it even harder for the scoop to pick up anything. This sent us back to the drawing board.

We continued on development with the chassis and electrical components, such as the coding and machine learning, but additionally brainstormed potential new parts we could use as our actuators. We decided to switch out our chassis; we were going with a small laser-cut square to minimize weight, but realized that this wasn’t going to work. We designed a chassis with holes in the bottom to screw our components into and attach our parts with something to fill the hole

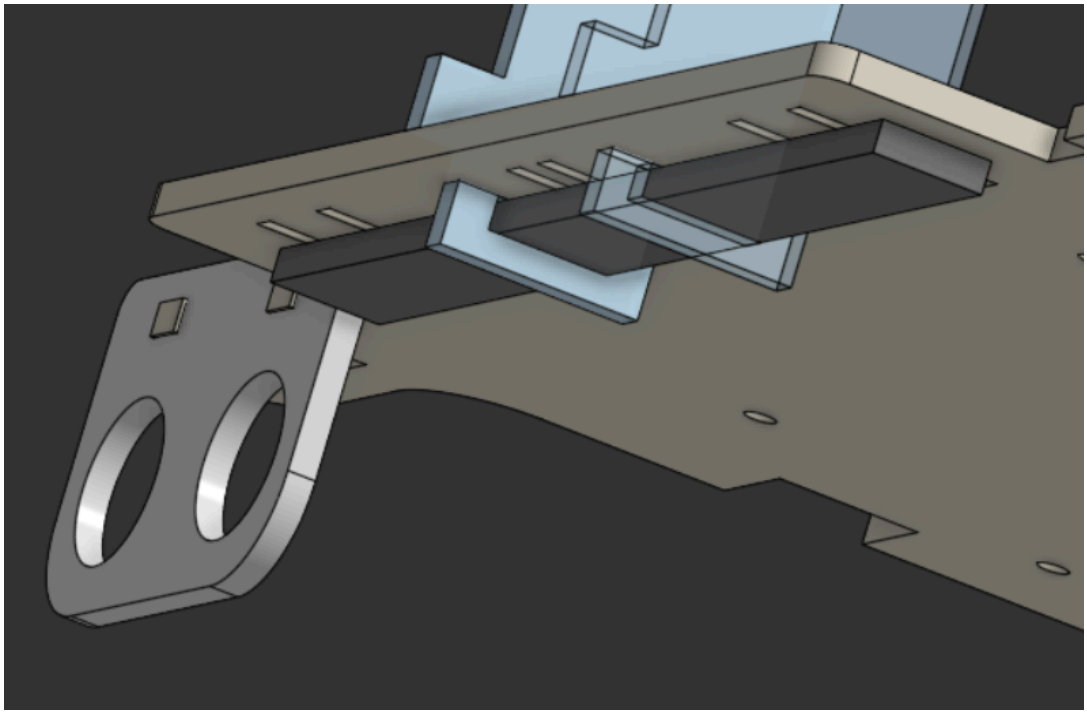
on the part. This chassis worked perfectly with our components, and we kept it throughout the rest of the course.



[2] Bottom view of our OTV chassis

Eventually, we came up with a design for our actuators. Instead of picking up the nonplantable substrate with a scoop, we decided to use a sticky substance. Our idea was to use a servo and to press down a pad with something sticky attached to the bottom to pick up the rocks. We went through many different types of sticky things, such as Command Strips, painters' tape, and electrical tape. Ultimately, we figured out that duct tape worked the best for picking up the rocks. W went through many different ideas for our other actuator, the planter, including the scooper and a type of vacuum. However, we landed on a ‘V’ shaped device that used a servo and a rack and pinion to press our seed and release our seed onto the orzo and then keep pressing it down until it gets deep enough. All of these were designed to be made out of wood, so everything could be laser cut and easy to remove and attach. We designed a battery holder and had a new design for the second layer of our OTV, where our Aruco marker is.

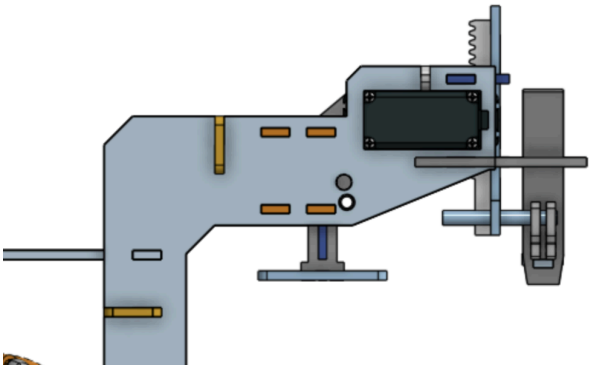
Finally, by milestone 6, we had our final design. Everything was made out of wood and able to be laser cut, besides some brackets for our wheels, the connectors between our wheels and our motors, and our self-designed gears, which were all 3D printed. Our mission actuator was one arm coming off of our chassis, so we were worried about it tipping forward, but to counteract this, we placed a little piece of wood to hold the structure together and tight.



[3] Mission support for our OTV

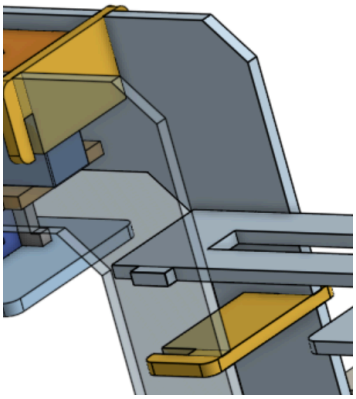
We used the second layer of our robot, where the Aruco marker lay, as a structural support for our arm and used a support beam under the chassis to hold the arm closer to our chassis. Anytime something did not fit right or needed to be redesigned, all we had to do was laser cut it, remove it easily from our OTV, and attach the new part in its place. This was because instead of bonding things to our OTV, we used holes to attach each part. This was very efficient and helped us keep moving along. A big problem we had was with the code; we had a hard time getting it to work. Every time we thought we had it working, another problem would show itself. One problem early on was that our machine learning only worked about 50% of the time. We decided to switch out our rocks to darker rocks so the camera would have an easier time detecting what was what. This worked perfectly, and the camera was able to give feedback on where the plantable substrate was and knew that the other plots had rocks in them. Near the end, our larger servo stopped working. We could hear that gears were turning on the inside, but it wouldn't turn the head. We decided to open up the servo with a screwdriver, take all the gears apart, and put it back together. After nearly an hour of trying to figure out around 10 gears, without any diagram to follow, we thought we put it back right, screwed it back together, and it was fixed! Soon after this, close to the end, when everything was set up, our wheels wouldn't work after the servos were connected. Lots of slightly altering the code and changing pins fixed this. A similar issue arose where after the wifi module was connected, the servos wouldn't work. Similar work fixed this. All the code and pins worked individually, but together, things broke. Fixing those last few problems took up the last few days leading up to the final test, meaning we couldn't test everything working together until the actual final test. We could not test everything together outside of class because we relied on the room's Jetson for Machine Learning.

Mission Details



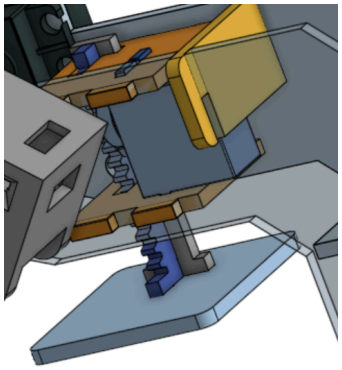
[4] CAD of Full Mission Apparatus

The mission apparatus consists of two pieces of laser-cut wood connected to the chassis by two slots on the front of the chassis.



[5] CAD of Supports

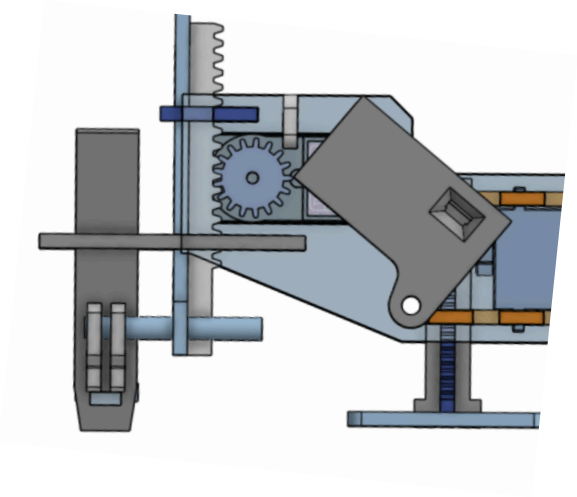
There are two laser–cut supports in between the cuts of wood to hold them together.



[6] CAD of Collecting Non-Plantable Substrate Actuator

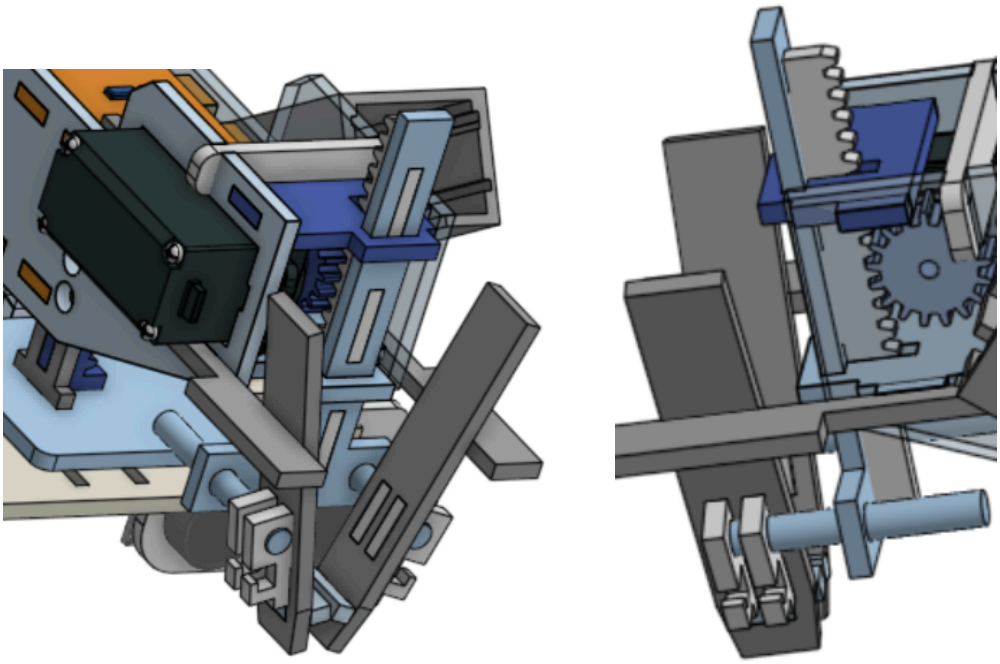
Further towards the front of the mission apparatus, a servo is kept in place by two horizontal laser-cut planks. The servo connects to a rack and pinion (the circular gear is 3d printed, and the long gear is laser-cut), and those are glued at the bottom to a flat laser-cut plank of wood. Before each run, a piece of duct tape is folded on itself to make a double-sided sticky

pad, and it is stuck to the bottom of the plank. This is for pebble collection.



[7] CAD of Camera Module

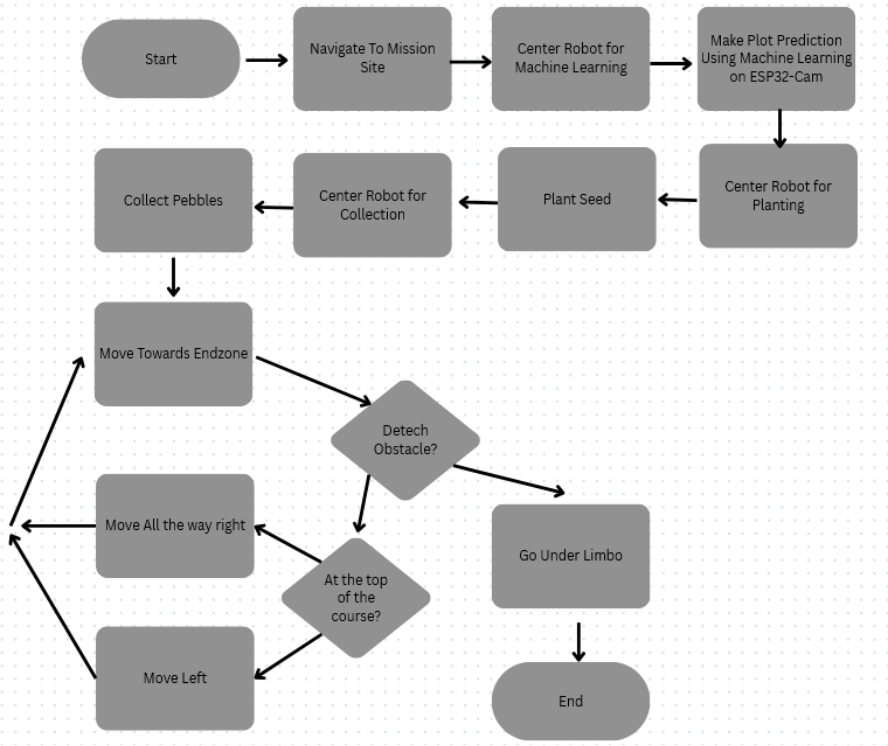
In front of that is the camera module, which is 3D printed. That is kept in place by a round wooden stick that is stuck through holes on the sides of the module and through holes in the mission apparatus. Tape is holding the camera at an angle so it can see all 4 pots from a bird's-eye view.



[8/9] CAD of Seed Planting Mechanism

At the front of the mission apparatus is the seed planting mechanism. A motor is put through a rectangular hole in the mission apparatus. The motor connects to a rack and pinion (the circular gear is 3d printed, and the long gear is laser-cut). The rest of the planting mechanism is all wooden laser-cut pieces. A rubber band goes at the bottom, wrapped a couple of times around the notches on either side of the mechanism to hold it together, which holds the seed in place

between the edges of the “pinchers.”



[10] This is the flowchart of how the robot’s mind will work.

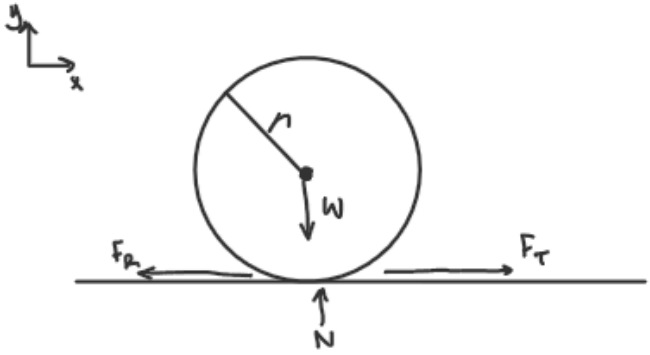
Through the provided resources, we have trained a program to identify samples of orzo and pebbles. Once it has identified the quadrant the orzo is in, it will send the information back to the ESP32-Camera, which travels to the Arduino Mega. The Arduino will take this information to plant the seed into the correct plot and sample pebbles from an adjacent one. Our OTV is powered by a 12V battery that is placed on the back of the OTV and is connected to a kill switch. More details about power and current distribution are outlined in the electronics section.

Propulsion Modeling

Configuration:
Wheels: 3 cm radius Mecanum Wheels
Wheel Motors: Servocity Mini-Econ Gear Motor 81RPM 12VDC,
Drive Train: Mecanum Drive Train
Steering: Mecanum Steering (Non-Steering)

Predicted Motor Behavior Calculations:

Torque Required by Design:



$$r = 3cm \qquad w = \frac{3kg}{4} * g = 7.35\ N$$

$$F_R = C * N = (0.004)(7.35N) = 0.0294N$$

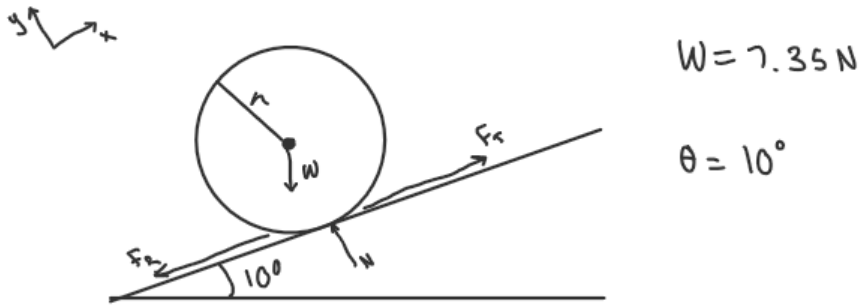
$$\Sigma F_x = 0\ ; F_T - F_R = 0 \rightarrow F_T = 0.0294N$$

$$\Sigma M_0 = 0\ ; (F_T)r\ -\ \tau_{req} = 0 \rightarrow\ \tau_{req} = 0.0294\ (0.03) = 0.000882\ N * m$$

[11] The calculations for the required torque to overcome rolling resistance

Angular Speed Calculation:

10 meters in 2.5 minutes: $\frac{10m}{2.5min} = 0.067\ \frac{m}{s} \rightarrow \frac{0.067\frac{m}{s}}{0.03m} \cdot \frac{60s}{1min} = 21.3\ RPM$



$$\Sigma F_y = 0\ ; N\ - \omega\ * \cos 10 \rightarrow N = \omega * \cos 10$$

$$F_R = C\ * N \rightarrow 0.004 * \omega \cos 10$$

$$\Sigma F_x = 0\ ;\ F_T - F_R - \omega \sin 10 = 0$$

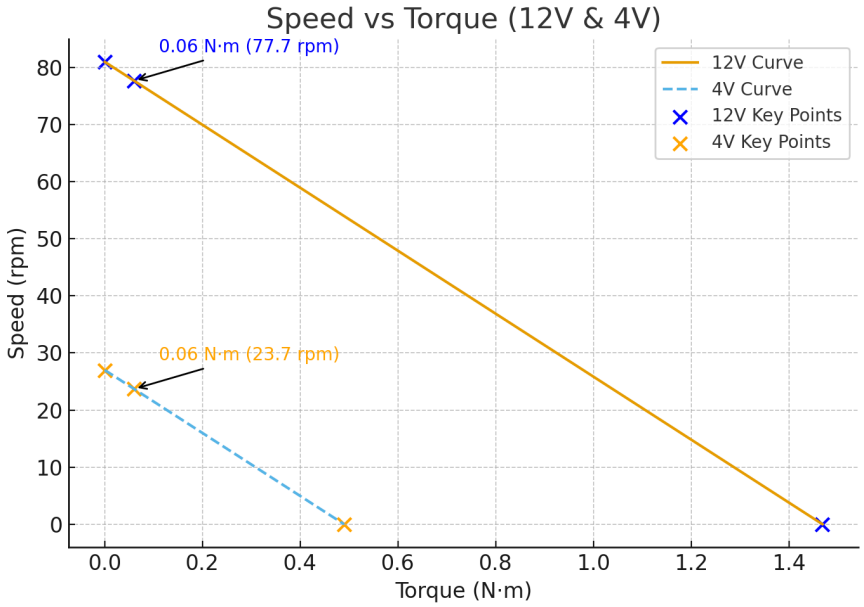
$$F_T = F_R + \omega \sin 10$$

$$\Sigma M_0 = 0\ ;\ F_T * r\ - \tau_{req} = 0$$

$$\tau_{req} = r\omega(\sin 10 + 0.004 \cos 10) = 0.04 * 7.35(\sin 10 + 0.004 \cos 10) = 0.0392N * m$$

[12] The image shows the force and torque analysis of a wheel on a 10° incline, calculating a required torque of 0.0392 N·m to overcome gravity and rolling resistance.

Motor characteristic graph:

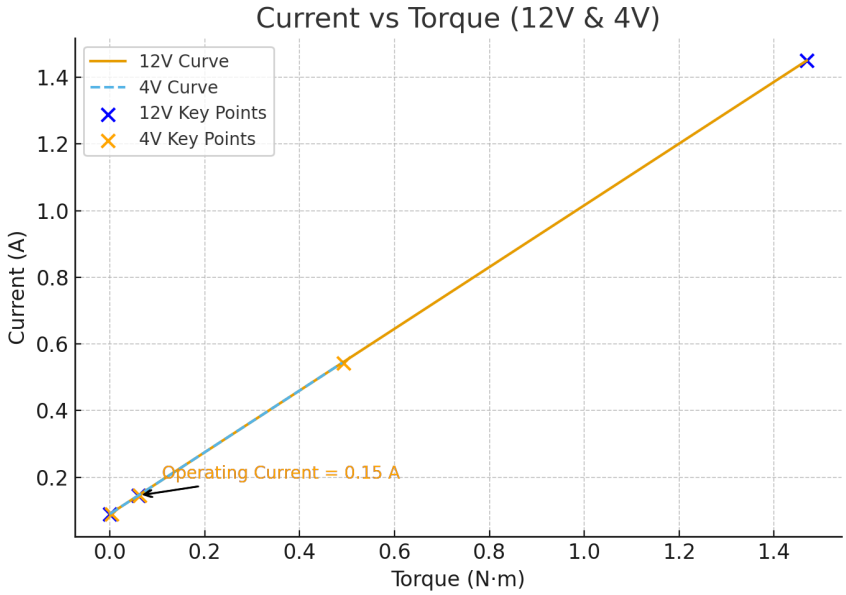


[13] The image shows a wheel on a 10° incline with forces acting on it and a motor speed-versus-torque graph comparing how the wheel's motor performs at 12 V and 4 V under different torque loads.

Operating linear speed:

$$v = \frac{2\pi r}{60} * \omega = \frac{2\pi(0.03m)}{60s} * 24RPM = 0.075m/s$$

Operating current graph:



[14] The graph above represents the proportional relationship between an increase in torque and an increase in current for both the 4V and 12V motors.

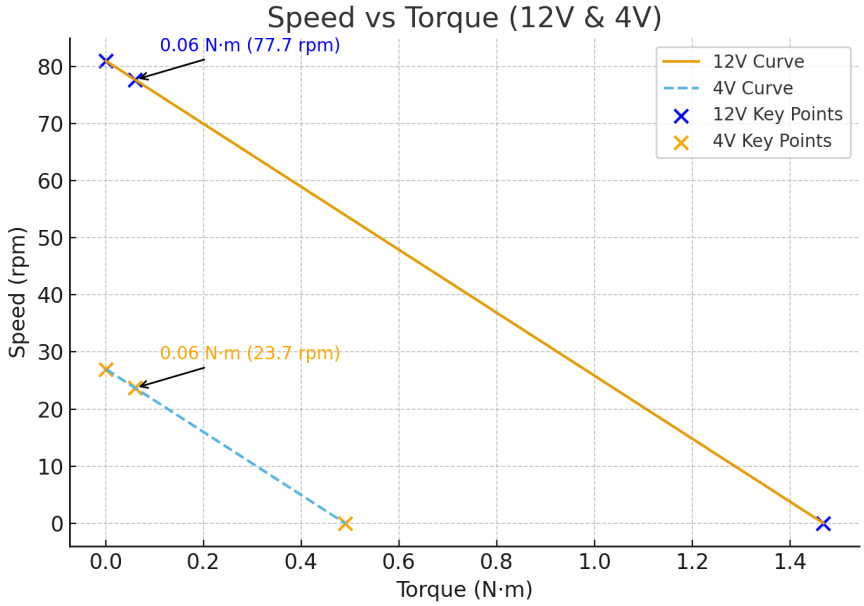
Actual Motor Behavior Calculations:

Operating Linear Speed:
As calculated in testing: **0.071m/s**

Operating Angular Speed:

$$\omega = \frac{v}{r} * \frac{60}{2\pi} = \frac{0.071\frac{m}{s}}{0.03m} * \frac{60}{2\pi} = 22.6\ RPM$$

Operating Torque:



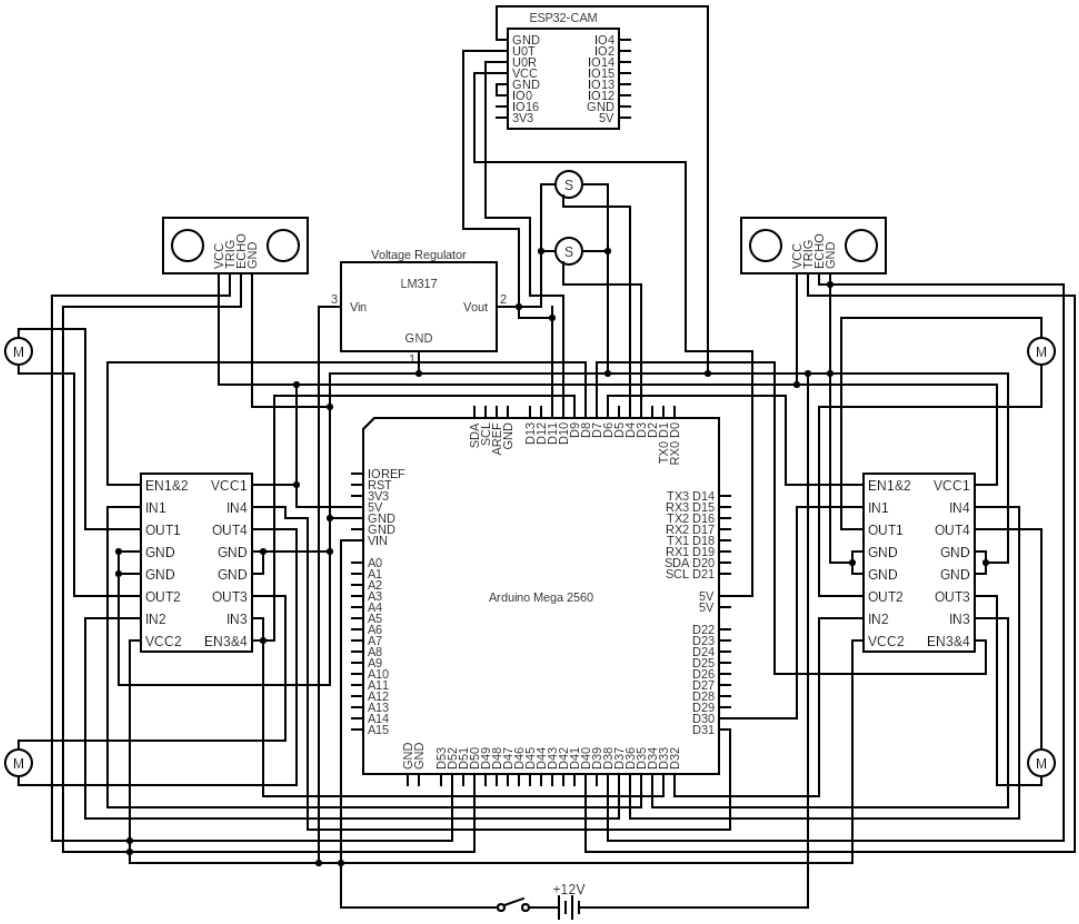
Based on graph 22.6 RPM corresponds to **1.05 N-m** of Torque

Torque Comparison:
Our actual operating torque exceeded our required and predicted torque values. This worked well in our favor as our OTV had minimal trouble with traction issues and traversing the uneven terrain of the arena. The main factor that likely contributed to this large difference was that the maximum RPM of our selected motors was almost 4 times that of what we were expecting to run our OTV at. When running at a fraction of the maximum speed, the torque increased as a result.

Electronics

Some of our design constraints are that we need a battery that can last for testing in the classroom. No Lithium or Lead Acid Batteries are allowed, and we require a 12V battery to power motors. We have to have enough digital pins on the Arduino Mega for all our components.

For our battery, we chose a 12V 2000mAh NiMH battery as it is rechargeable and has a runtime of 0.79 hours with our current robot setup, which will be directly connected to a kill switch. It has a max continuous discharge current of 2A and a max recharge rate of 1A maximum; >= 500 cycles; 255g; 73.5x53.5x30.5mm (LxWxT)



[15] A circuit diagram for our OTV electronics.

We also have two ultrasonic sensors on the front of our robot; these are used to detect obstacles, center the OTV, and distance ourselves from the mission site appropriately. In addition, we have an ESP32-Camera to transmit a video data stream to the camera above the mission site. This camera has a built-in wifi module for our team to use. We also have a servo motor controlling our planting mechanism on the front of our robot. To have the appropriate pins for our robot, we will be using an Arduino Mega. We will have two H-Bridges to control the robot's 4 motors, which will be placed on the robot next to the Arduino. The ESP32 wifi module and camera's full power was only used for a few seconds, and the servos needed negligible extra

power for only a few seconds, so the total uses the minimum draw for both.

Current Draw Calculations

- Arduino Mega: 50mA
- ESP32 Wifi Module and Camera: 90mA, up to 500mA
- Ultrasonic Sensors: 15mA x 2 = 30mA
- Motors: 320mAx 4 = 1280mA
- Servo FS90R: 100mA
- Servo MG996R: 500mA
- Total: 2050mA

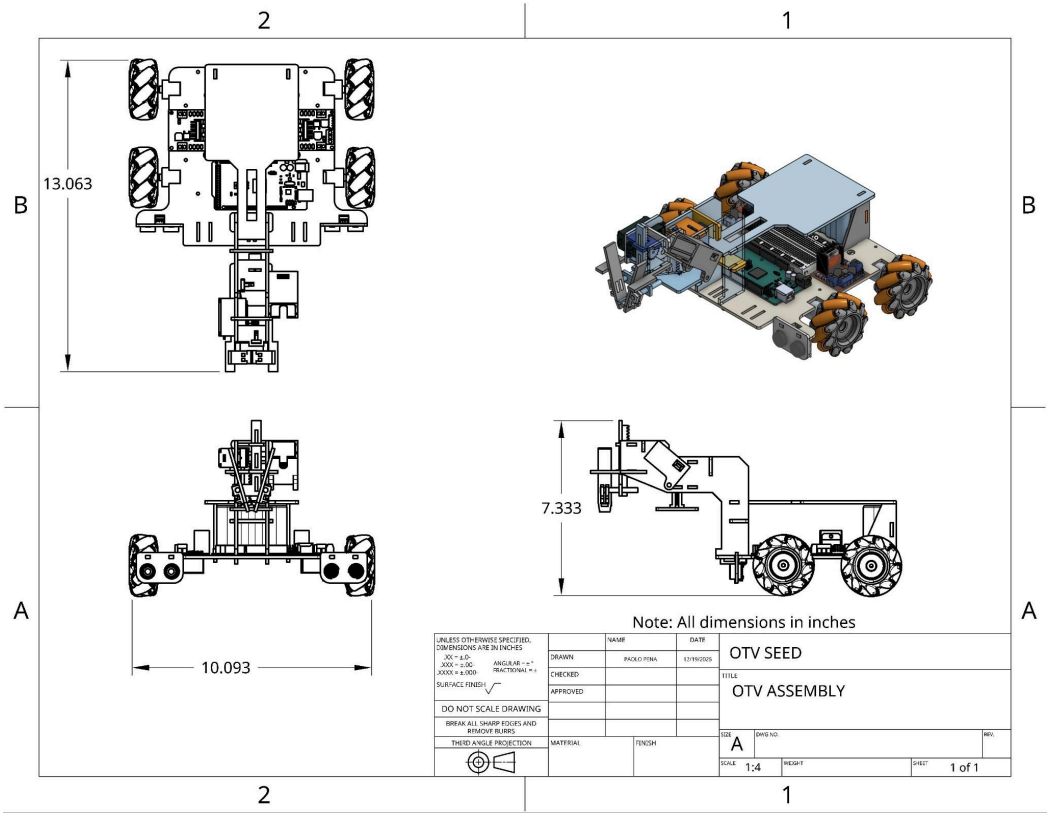
Run Time Calculations

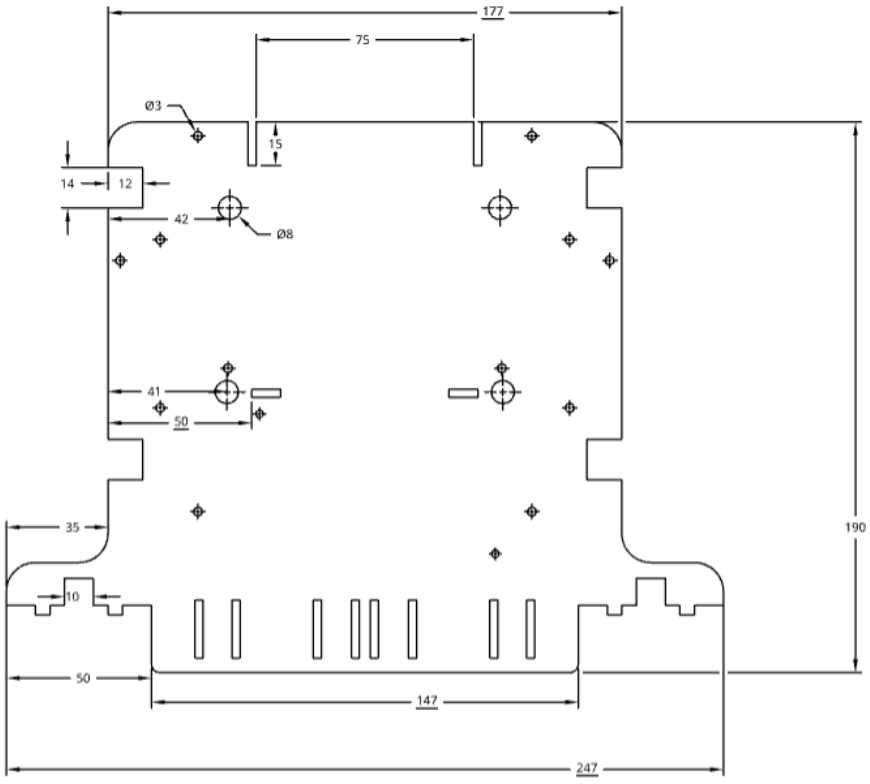
- Battery Capacity: 2Ah
- Total Current Draw: 2050 mA
- $\frac{2Ah}{2.05A} = 0.98\ hours$

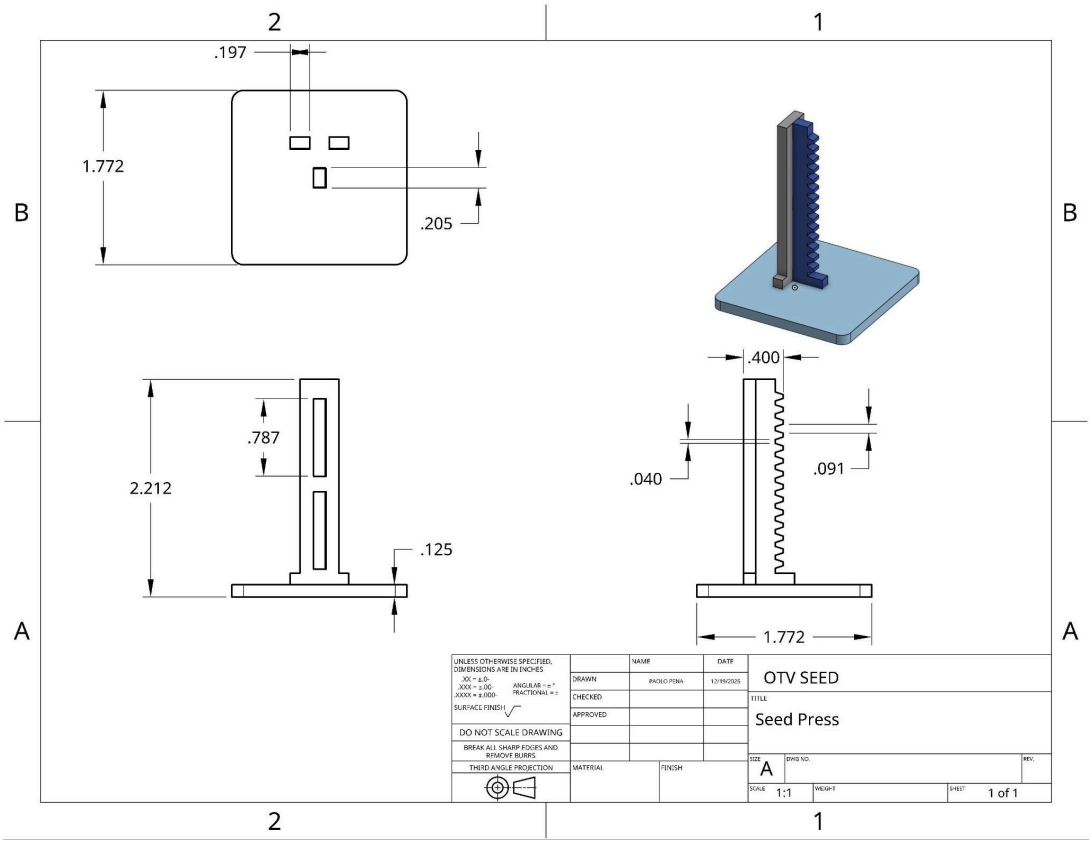
The current draw does exceed the maximum discharge rate of our battery; however, not by much. As mentioned earlier, the 12V battery will connect directly to the H-Bridge's motor inputs, the input voltage for the voltage regulator, and to the Vin pin on the Arduino Mega. From there, the servos will be powered from the other side of the voltage regulator, the ESP32-Cam will be powered directly from a 5V pin on the Arduino, and every other component will be connected to a 5V rail on a breadboard (which is connected to the regulated 5V output of the H-Bridges).

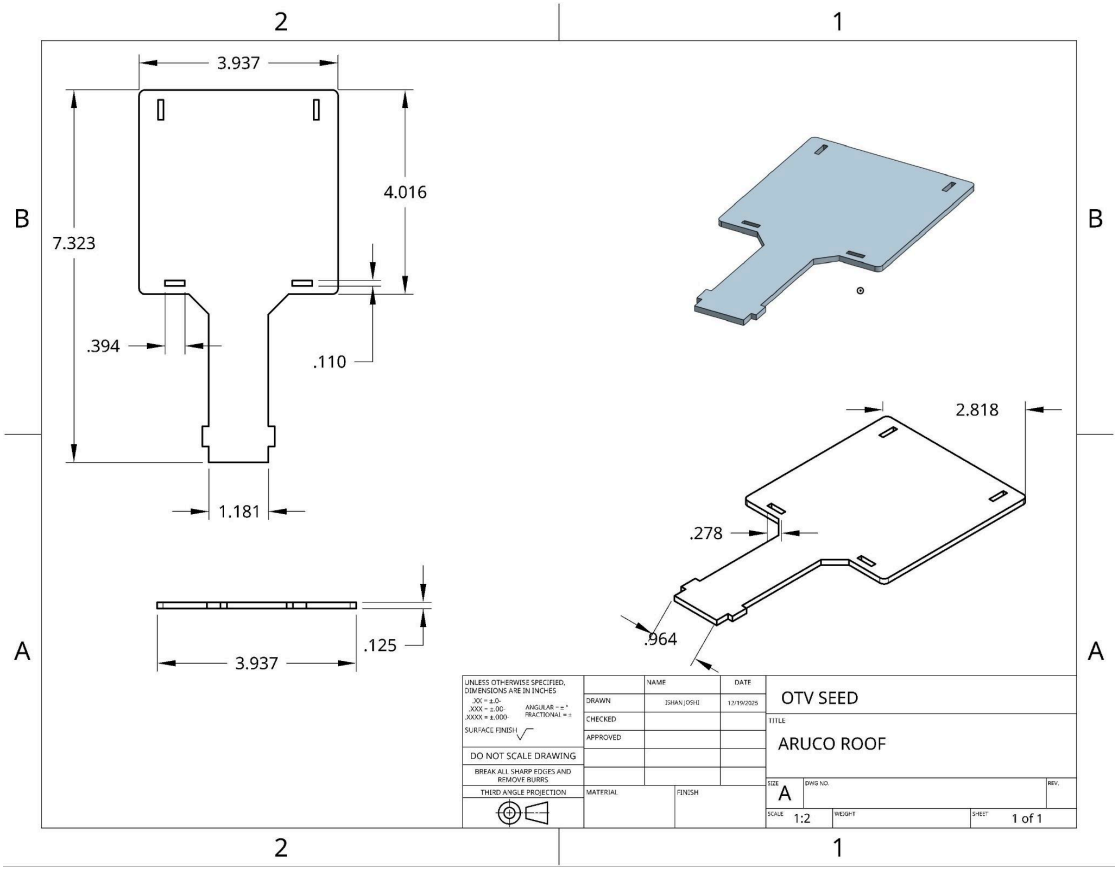
D3	Servo1 Signal	D11	U0T - ESP Cam	D36	H-Bridge2-Input4
D4	Servo2 Signal	D30	H-Bridge2-Input2	D37	H-Bridge1-Input4
D6	Enable A 1	D31	H-Bridge1-Input1	D38	Echo Pin 2
D7	Enable B 1	D32	H-Bridge2-Input1	D40	Trig Pin 2
D8	Enable A 2	D33	H-Bridge1-Input2	D50	Trig Pin 1
D9	Enable B 2	D34	H-Bridge2-Input3	D52	Echo Pin 1
D10	U0R - ESP Cam	D35	H-Bridge1-Input3		15 open pins

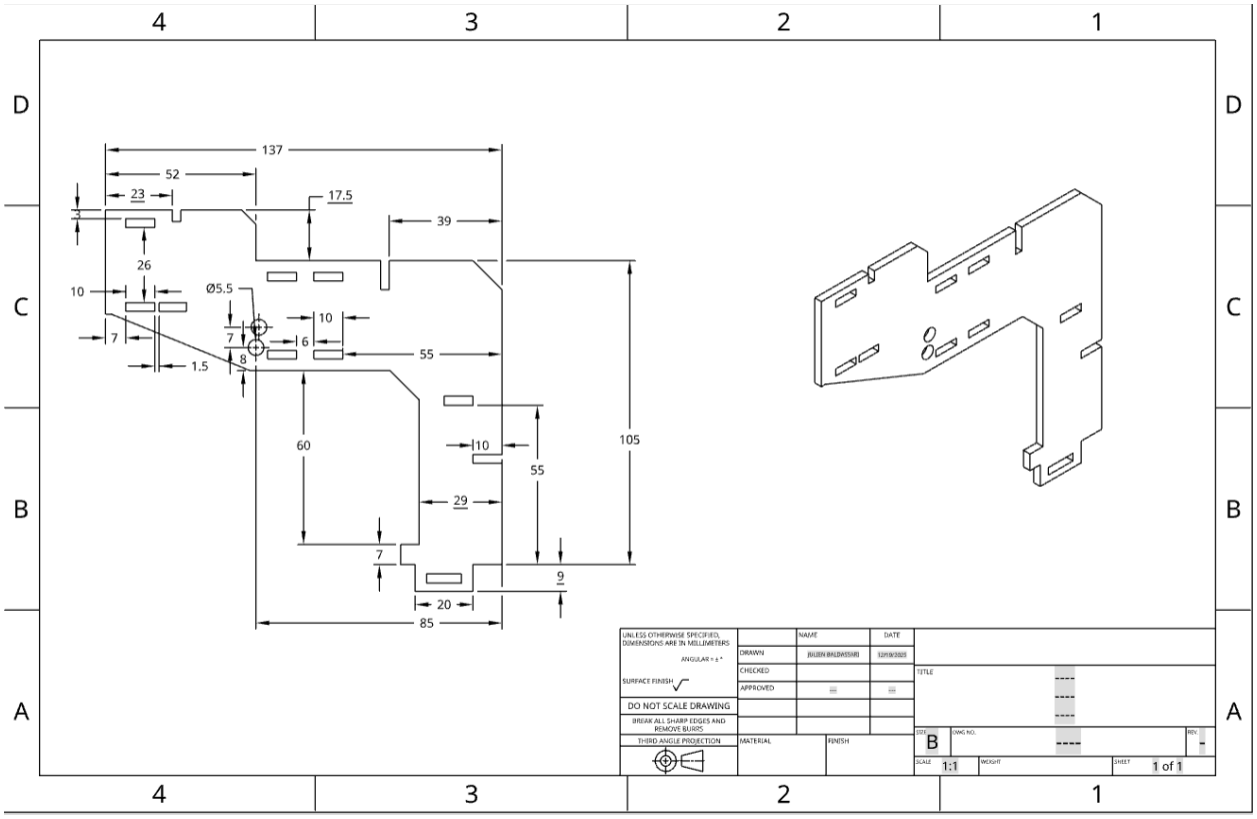
Engineering Drawings

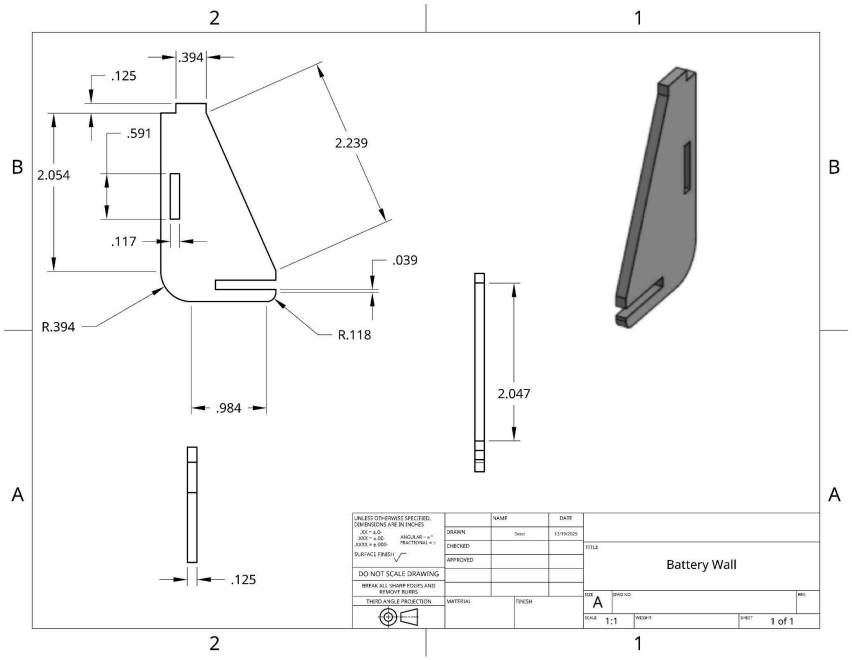


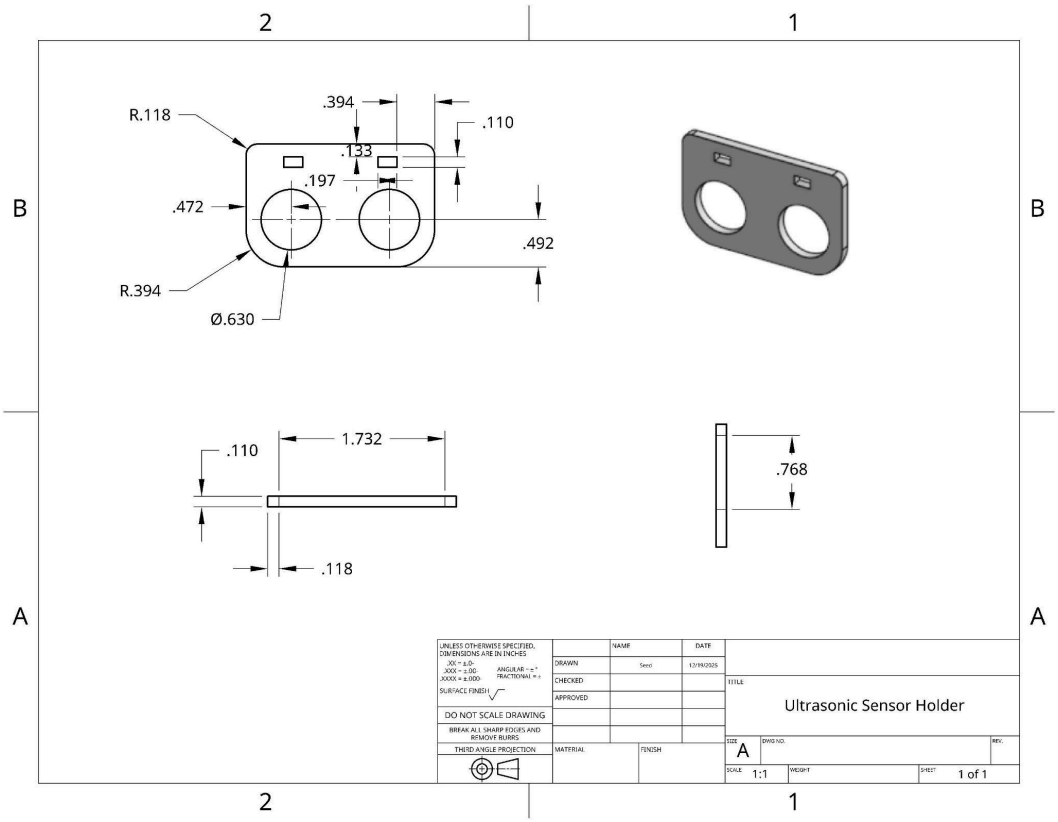



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[16, 17, 18, 19, 20, 21, 22] Engineering drawings from our CAD, all with measurements

Sustainability

Ansys
GRANTA EDUPACK

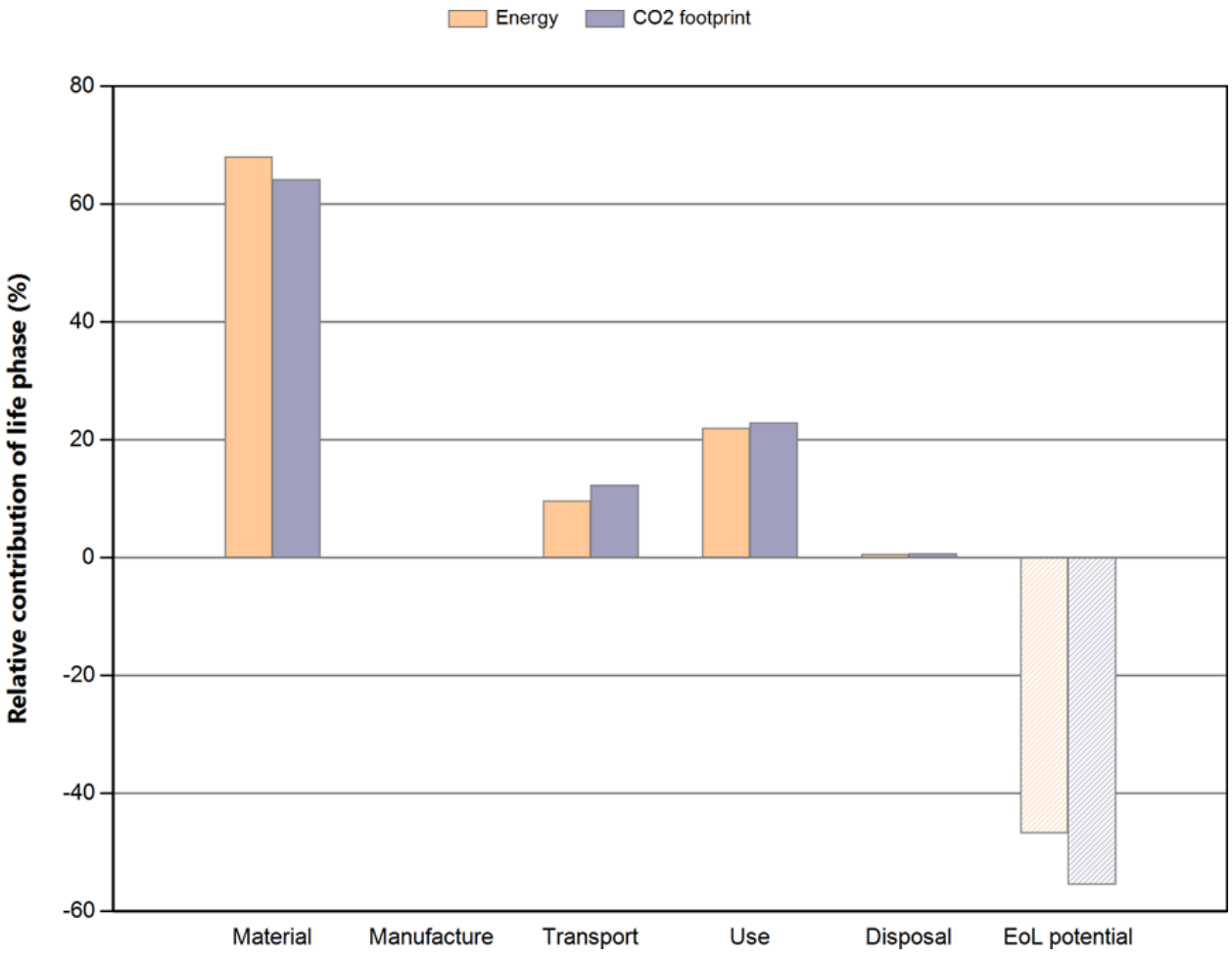
Eco Audit Report

Product name **OTV_DirtDivas**

Country of use **United States**

Product life (years) **0.25**

Summary:



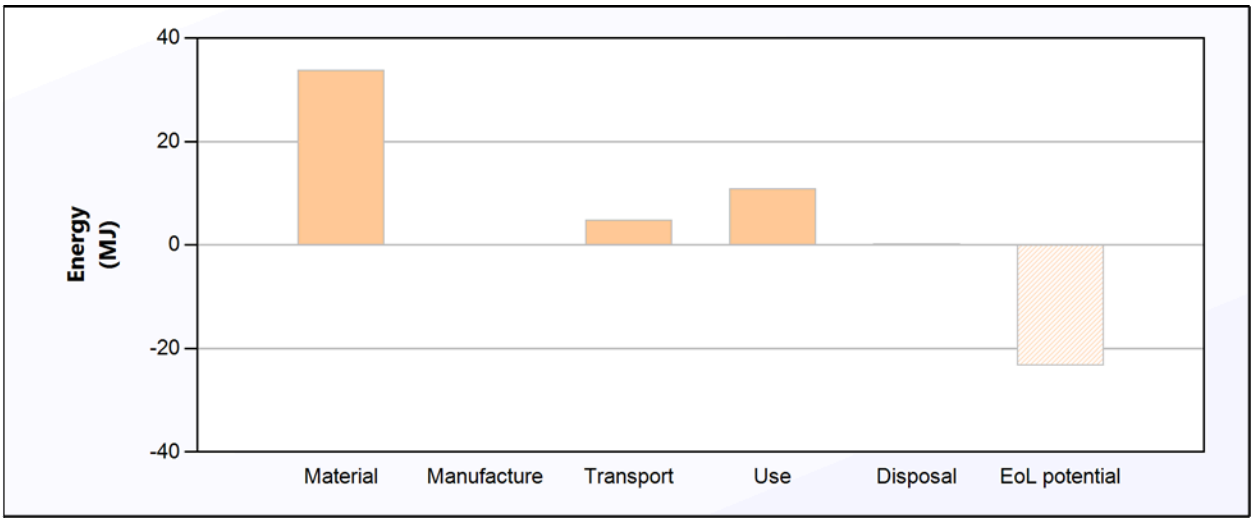
Energy details CO2 footprint details

Phase	Energy (MJ)	Energy (%)	CO2 footprint (kg)	CO2 footpr

				int (%)
Material	**33.8**	**67.9**	**1.8**	**64.2**
Manufacture	**0**	**0.0**	**0**	**0.0**
Transport	**4.79**	**9.6**	**0.345**	**12.3**
Use	**10.9**	**21.9**	**0.641**	**22.9**
Disposal	**0.255**	**0.5**	**0.0179**	**0.6**
Total (for first life)	**49.8**	**100**	**2.8**	**100**
End of life potential	**-23.3**		**-1.56**	

Energy Analysis

Summary



	Energy (MJ/year)
Equivalent annual environmental burden (averaged over 0.25 year product life):	**199**

Detailed breakdown of individual life phases

Material:

Summary

Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass (kg)	Energy (MJ)	%
Arduino Mega	Printed circuit board assembly	Virgin (0%)	0.035	1	0.035	4.5	13.4
Motor	Fan	Reused part	0.043	4	0.17	0	0.0
Screw	Stainless steel, austenitic, AISI 202, annealed	Virgin (0%)	0.001	6	0.006	0.35	1.0
Rechargeable NiMH Battery	NiMH, rechargeable battery (for laptops)	Reused part	0.25	1	0.25	0	0.0
Mechanum Wheel Frame	PA66 (30-33% glass fiber)	Reused part	0.035	4	0.14	0	0.0
Mechanum Wheel Rollers	TPU (Ester, aromatic, Shore A70)	Reused part	0.015	4	0.06	0	0.0
Motor Driver	Printed circuit board assembly	Virgin (0%)	0.03	2	0.06	7.8	23.0
Breadboard Body	ABS (injection molding, platable)	Virgin (0%)	0.045	1	0.045	4.3	12.8
Breadboard Internal Contacts	Bronze, CuSn4, C51000, hard (5% phosphor bronze)	Virgin (0%)	0.005	1	0.005	0.51	1.5
Wires	Cable	Virgin (0%)	0.001	45	0.045	4.1	12.1
Micro Servo Shell	ABS (injection molding, platable)	Reused part	0.0055	1	0.0055	0	0.0
Micro Servo Components	Printed circuit board assembly	Reused part	0.0025	1	0.0025	0	0.0

High Torque Servo Shell	ABS (injection molding, platable)	Reused part	0.0075	1	0.0075	0	0.0
High Torque Servo Components	Printed circuit board assembly	Reused part	0.0045	1	0.0045	0	0.0
HC-SR04 Ultrasonic Secor	Epoxy/S-glass fiber, UD prepreg, QI lay-up	Virgin (0%)	0.009	2	0.018	2	6.0
Birch Sheets	Birch (betula alleghaniensis) (l)	Virgin (0%)	0.39	1	0.39	8.6	25.5
Wheel Hub	PLA (10% glass fiber)	Virgin (0%)	0.004	4	0.016	0.78	2.3
Motor Mounts	PLA (10% glass fiber)	Virgin (0%)	0.0035	4	0.014	0.69	2.0
Servo Gears	PLA (10% glass fiber)	Virgin (0%)	0.001	2	0.002	0.098	0.3
Total				86	1.3	34	100

*Typical: Includes 'recycle fraction in current supply'

***User-defined material

Manufacture:

Summary

Component	Process	Amount processed	Energy (MJ)	%
Total				100

Transport:

Breakdown by transport stage

Summary

Stage name	Transport type	Distance (km)	Energy (MJ)	%
Arduino Mega	32 tonne (4 axle) truck	8e+02	0.96	20.0
Motor Drivers	32 tonne (4 axle) truck	8e+02	0.96	20.0
HC-SR04 Ultrasonic Sesors	32 tonne (4 axle) truck	8e+02	0.96	20.0
Breadboard	32 tonne (4 axle) truck	8e+02	0.96	20.0
Wires	32 tonne (4 axle) truck	8e+02	0.96	20.0
Total		4e+03	4.8	100

Breakdown by components

Component	Mass (kg)	Energy (MJ)	%
Arduino Mega	0.035	0.13	2.7
Motor	0.17	0.65	13.5
Screw	0.006	0.023	0.5
Rechargeable NiMH Battery	0.25	0.94	19.6
Mecanum Wheel Frame	0.14	0.53	11.0
Mecanum Wheel Rollers	0.06	0.23	4.7
Motor Driver	0.06	0.23	4.7

Breadboard Body	**0.045**	**0.17**	**3.5**
Breadboard Internal Contacts	**0.005**	**0.019**	**0.4**
Wires	**0.045**	**0.17**	**3.5**
Micro Servo Shell	**0.0055**	**0.021**	**0.4**
Micro Servo Components	**0.0025**	**0.0094**	**0.2**
High Torque Servo Shell	**0.0075**	**0.028**	**0.6**
High Torque Servo Components	**0.0045**	**0.017**	**0.4**
HC-SR04 Ultrasonic Sensor	**0.018**	**0.068**	**1.4**
Birch Sheets	**0.39**	**1.5**	**30.4**
Wheel Hub	**0.016**	**0.06**	**1.3**
Motor Mounts	**0.014**	**0.053**	**1.1**
Servo Gears	**0.002**	**0.0075**	**0.2**
Total	**1.3**	**4.8**	**100**

Use:

Summary

Static mode

Energy input and output type	**Electric to mechanical (electric motors)**
Country of use	**United States**
Power rating (W)	**24**
Usage (hours per day)	**2**
Usage (days per year)	**1e+02**
Product life (years)	**0.25**

Relative contribution of static and mobile modes

Mode	Energy (MJ)	%
Static	**11**	**100.0**
Mobile	**0**	
Total	**11**	**100**

Disposal:

Component	End of life option	Energy (MJ)	%
Arduino Mega	Reuse	0.007	2.7
Motor	Reuse	0.034	13.5
Screw	Landfill	0.0012	0.5
Rechargeable NiMH Battery	Reuse	0.05	19.6
Mecanum Wheel Frame	Reuse	0.028	11.0
Mecanum Wheel Rollers	Reuse	0.012	4.7
Motor Driver	Reuse	0.012	4.7
Breadboard Body	Reuse	0.009	3.5
Breadboard Internal Contacts	Reuse	0.001	0.4
Wires	Reuse	0.009	3.5
Micro Servo Shell	Reuse	0.0011	0.4
Micro Servo Components	Reuse	0.0005	0.2
High Torque Servo Shell	Reuse	0.0015	0.6
High Torque Servo Components	Reuse	0.0009	0.4
HC-SR04 Ultrasonic Sensor	Reuse	0.0036	1.4
Birch Sheets	Landfill	0.077	30.4
Wheel Hub	Landfill	0.0032	1.3
Motor Mounts	Landfill	0.0028	1.1
Servo Gears	Landfill	0.0004	0.2
Total		0.26	100

EoL potential:

Summary

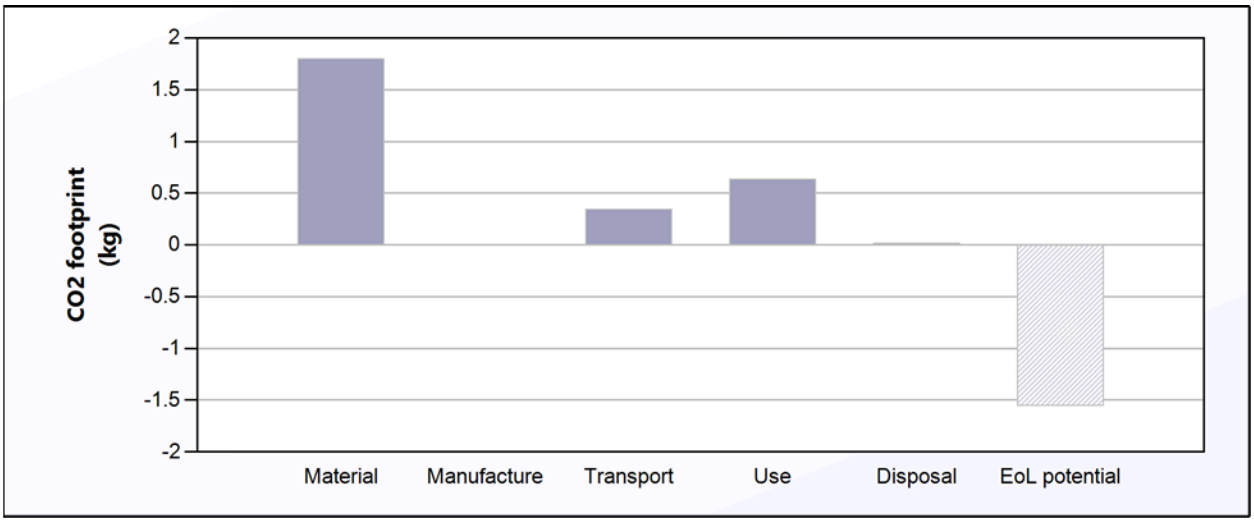
Component	End of life option	Energy (MJ)	%
Arduino Mega	Reuse	-4.5	19.5
Motor	Reuse	0	0.0
Screw	Landfill	0	0.0
Rechargeable NiMH Battery	Reuse	0	0.0
Mechanum Wheel Frame	Reuse	0	0.0
Mechanum Wheel Rollers	Reuse	0	0.0
Motor Driver	Reuse	-7.8	33.4
Breadboard Body	Reuse	-4.3	18.6
Breadboard Internal Contacts	Reuse	-0.51	2.2
Wires	Reuse	-4.1	17.6
Micro Servo Shell	Reuse	0	0.0
Micro Servo Components	Reuse	0	0.0
High Torque Servo Shell	Reuse	0	0.0
High Torque Servo Components	Reuse	0	0.0
HC-SR04 Ultrasonic Secor	Reuse	-2	8.7
Birch Sheets	Landfill	0	0.0
Wheel Hub	Landfill	0	0.0
Motor Mounts	Landfill	0	0.0
Servo Gears	Landfill	0	0.0
Total		-23	100

Notes:

Summary

CO2 Footprint Analysis

Summary



	CO2 (kg/year)
Equivalent annual environmental burden (averaged over 0.25 year product life):	11.2

Detailed breakdown of individual life phases

Material:

Summary

Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass (kg)	CO2 footprint (kg)	%
Arduino Mega	Printed circuit board assembly	Virgin (0%)	0.035	1	0.035	0.34	18.9
Motor	Fan	Reused part	0.043	4	0.17	0	0.0
Screw	Stainless steel, austenitic, AISI 202, annealed	Virgin (0%)	0.001	6	0.006	0.031	1.7

Rechargeable NiMH Battery	NiMH, rechargeable battery (for laptops)	Reused part	0.25	1	0.25	0	0.0
Mecanum Wheel Frame	PA66 (30-33% glass fiber)	Reused part	0.035	4	0.14	0	0.0
Mecanum Wheel Rollers	TPU (Ester, aromatic, Shore A70)	Reused part	0.015	4	0.06	0	0.0
Motor Driver	Printed circuit board assembly	Virgin (0%)	0.03	2	0.06	0.58	32.4
Breadboard Body	ABS (injection molding, platable)	Virgin (0%)	0.045	1	0.045	0.17	9.2
Breadboard Internal Contacts	Bronze, CuSn4, C51000, hard (5% phosphor bronze)	Virgin (0%)	0.005	1	0.005	0.035	2.0
Wires	Cable	Virgin (0%)	0.001	45	0.045	0.31	17.0
Micro Servo Shell	ABS (injection molding, platable)	Reused part	0.0055	1	0.0055	0	0.0
Micro Servo Components	Printed circuit board assembly	Reused part	0.0025	1	0.0025	0	0.0
High Torque Servo Shell	ABS (injection molding, platable)	Reused part	0.0075	1	0.0075	0	0.0
High Torque Servo Components	Printed circuit board assembly	Reused part	0.0045	1	0.0045	0	0.0
HC-SR04 Ultrasonic Secor	Epoxy/S-glass fiber, UD prepreg, QI lay-up	Virgin (0%)	0.009	2	0.018	0.12	6.9
Birch Sheets	Birch (betula alleghaniensis) (l)	Virgin (0%)	0.39	1	0.39	0.13	7.1

Wheel Hub	PLA (10% glass fiber)	Virgin (0%)	0.004	4	0.016	0.043	2.4
Motor Mounts	PLA (10% glass fiber)	Virgin (0%)	0.0035	4	0.014	0.038	2.1
Servo Gears	PLA (10% glass fiber)	Virgin (0%)	0.001	2	0.002	0.0054	0.3
Total				86	1.3	1.8	100

*Typical: Includes 'recycle fraction in current supply'

***User-defined material

Manufacture:

Summary

Component	Process	Amount processed	CO2 footprint (kg)	%
Total				100

Transport:

Breakdown by transport stage

Summary

Stage name	Transport type	Distance (km)	CO2 footprint (kg)	%
Arduino Mega	32 tonne (4 axle) truck	8e+02	0.069	20.0
Motor Drivers	32 tonne (4 axle) truck	8e+02	0.069	20.0
HC-SR04 Ultrasonic Sensors	32 tonne (4 axle) truck	8e+02	0.069	20.0

Breadboard	**32 tonne (4 axle) truck**	**8e+02**	**0.069**	**20.0**
Wires	**32 tonne (4 axle) truck**	**8e+02**	**0.069**	**20.0**
Total		**4e+03**	**0.35**	**100**

Breakdown by components

Component	Mass (kg)	CO2 footprint (kg)	%
Arduino Mega	**0.035**	**0.0095**	**2.7**
Motor	**0.17**	**0.047**	**13.5**
Screw	**0.006**	**0.0016**	**0.5**
Rechargeable NiMH Battery	**0.25**	**0.068**	**19.6**
Mecanum Wheel Frame	**0.14**	**0.038**	**11.0**
Mecanum Wheel Rollers	**0.06**	**0.016**	**4.7**
Motor Driver	**0.06**	**0.016**	**4.7**
Breadboard Body	**0.045**	**0.012**	**3.5**
Breadboard Internal Contacts	**0.005**	**0.0014**	**0.4**
Wires	**0.045**	**0.012**	**3.5**
Micro Servo Shell	**0.0055**	**0.0015**	**0.4**
Micro Servo Components	**0.0025**	**0.00068**	**0.2**
High Torque Servo Shell	**0.0075**	**0.002**	**0.6**

High Torque Servo Components	0.0045	0.0012	0.4
HC-SR04 Ultrasonic Sensor	0.018	0.0049	1.4
Birch Sheets	0.39	0.1	30.4
Wheel Hub	0.016	0.0043	1.3
Motor Mounts	0.014	0.0038	1.1
Servo Gears	0.002	0.00054	0.2
Total	1.3	0.35	100

Use:

Summary

Static mode

Energy input and output type	Electric to mechanical (electric motors)
Country of use	United States
Power rating (W)	24
Usage (hours per day)	2
Usage (days per year)	1e+02
Product life (years)	0.25

Relative contribution of static and mobile modes

Mode	CO2 footprint (kg)	%
Static	0.64	100.0
Mobile	0	
Total	0.64	100

Disposal:

Component	End of life option	CO2 footprint (kg)	%
Arduino Mega	Reuse	0.00049	2.7
Motor	Reuse	0.0024	13.5
Screw	Landfill	8.4e-05	0.5
Rechargeable NiMH Battery	Reuse	0.0035	19.6
Mecanum Wheel Frame	Reuse	0.002	11.0
Mecanum Wheel Rollers	Reuse	0.00084	4.7
Motor Driver	Reuse	0.00084	4.7
Breadboard Body	Reuse	0.00063	3.5
Breadboard Internal Contacts	Reuse	7e-05	0.4
Wires	Reuse	0.00063	3.5
Micro Servo Shell	Reuse	7.7e-05	0.4
Micro Servo Components	Reuse	3.5e-05	0.2
High Torque Servo Shell	Reuse	0.00011	0.6

High Torque Servo Components	Reuse	6.3e-05	0.4
HC-SR04 Ultrasonic Sensor	Reuse	0.00025	1.4
Birch Sheets	Landfill	0.0054	30.4
Wheel Hub	Landfill	0.00022	1.3
Motor Mounts	Landfill	0.0002	1.1
Servo Gears	Landfill	2.8e-05	0.2
Total		0.018	100

EoL potential:

Summary

Component	End of life option	CO2 footprint (kg)	%
Arduino Mega	Reuse	-0.34	21.9
Motor	Reuse	0	0.0
Screw	Landfill	0	0.0
Rechargeable NiMH Battery	Reuse	0	0.0
Mecanum Wheel Frame	Reuse	0	0.0
Mecanum Wheel Rollers	Reuse	0	0.0
Motor Driver	Reuse	-0.58	37.5
Breadboard Body	Reuse	-0.17	10.7
Breadboard Internal Contacts	Reuse	-0.035	2.3
Wires	Reuse	-0.31	19.7
Micro Servo Shell	Reuse	0	0.0
Micro Servo Components	Reuse	0	0.0
High Torque Servo Shell	Reuse	0	0.0
High Torque Servo Components	Reuse	0	0.0
HC-SR04 Ultrasonic Sensor	Reuse	-0.12	8.0

Birch Sheets	**Landfill**	**0**	**0.0**
Wheel Hub	**Landfill**	**0**	**0.0**
Motor Mounts	**Landfill**	**0**	**0.0**
Servo Gears	**Landfill**	**0**	**0.0**
Total		**-1.6**	**100**

Notes:

Summary